Study of Different types of Network Cables.

Exp.no : 2
Date : 30/07/2026

Aim:

To study the different types of Network cables.

a:) Understand different types of network cable.

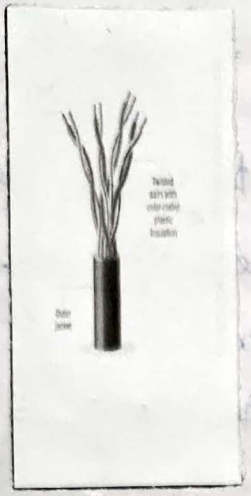
Different type of cables used in networking are

1. Unshielded Twisted pair (UTP) cable
2. Shielded Twisted pair (STP) cable
3. coaxial cable
4. Fibre optic cable.

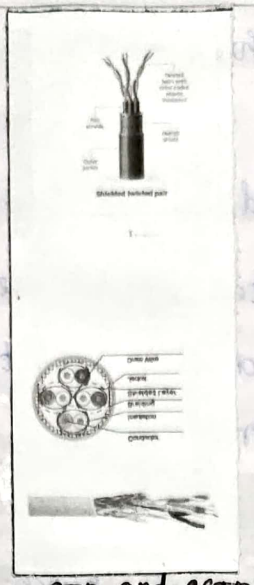
cable type	Category	Maximum Data Transmission	Advantages / Disadvantages	Application/ Use
UTP	Category 3	10 bps	Advantages :- → cheaper in cost → Easy to install as they have a smaller overall diameter Disadvantages :- → More prone to (EMI) Electromagnetic interference and noise	→ 10Base-T ethernet → Fast Ethernet, Gigabit Ethernet → ~~Fast~~
	category 5	Up to 100 Mbps		
	Category 5e	1 Gbps		
STP	Category6, 6a	10 Gbps	Advantages :- → shielded. → Faster than UTP. → Less susceptible to noise and	→ Gigabit Ethernet, 10G Ethernet (55m) widely used in data centres.

Scanned with OKEN Scanner

SSTP	category 7 ~~SSTP~~	10 Gbps ~~Category 7~~	interference disadvantages: → Expensive → Greater installation effort	→ Gigabit ethernet, 10G Ethernet (100 m)
Coaxial Cable	RG-6 RG-59 RG-11	10 - 100 Mbps	Advantages :- → High bandwidth → Immune to interference → Low loss bandwidth → Versatile Disadvantages :- → Limited distance → cost → size is bulky	→ speed of signal is 500m → Television network → High speed internet connections.
fibre optics Cable	Single mode Multi mode	100 Gbps	Advantages :- → High speed → High bandwidth → High security → Long distance disadvantages :- → Expensive → Requires skilled Installers	→ Maximum distance of fibre optics cable is around 100 meters.



UTP



STP and SSTP



Coaxial cable



fibre optics cable

b) Make your own Ethernet Cross-over cable / straight cable

Tools and parts needed :-

→ Ethernet cabling. CAT5e is certified for gigabit support, but CAT5 cabling works as well, just over shorter distances.

→ A crimping tool. This is an all-in one networking tool shaped to push down the pins in the plug and strip and cut the shielding off the cables.

→ Two RJ45 plugs.

→ optional two plug shields.

Step 1 :

To start construction of the device, begin by threading shields onto the cable.

Step 2 :

Next, strip approximately 1.5 cm of cable shielding from both ends. The crimping tool has a round

area to complete this task.

Step 3:

After, you will need to untangle the wires; there should be four "twisted pairs". Referencing back to the sheet, arrange them from top to bottom. One end should be in arrangement A and the other in B.

Step 4:

Once the order is correct, bunch them together in a line, and if there are any that stick out farther than others, snip them back to create an even level. The difficult aspect is placing these into the RJ45 plug without messing up the order. To do so, hold the plug with the clip side facing away from you and have the gold pins facing towards you, as shown.

Step 5:

Next, push the cable right in. The notch at the end of the plug needs to be just over the cable shielding and if it isn't, that means that you stripped off too much shielding. Simply snip the cables back a little more.

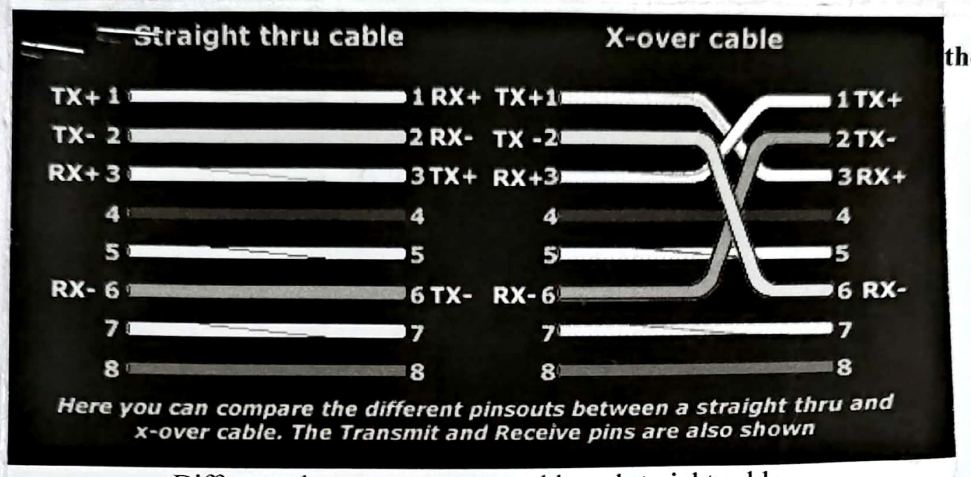
Step 6:

After the wires are securely sitting inside the plug, insert it into the crimping tool and push down. It should be shaped correctly, but pushing too hard can crack the fragile plastic plug.

Step 7:

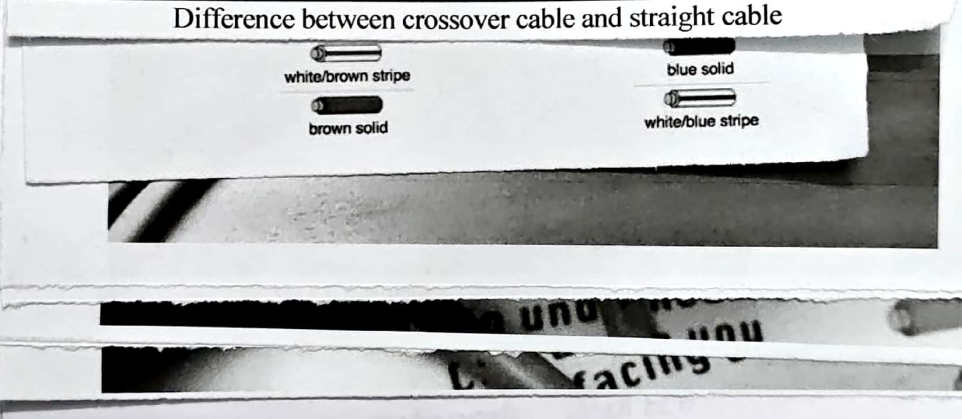
Lastly, repeat for the other end using

diagram B (to make a crossover cables) / using diagram A (to make a straight through cable). To test it, plug it in and attempt to connect two devices directly.



Here you can compare the different pinsouts between a straight thru and x-over cable. The Transmit and Receive pins are also shown

Difference between crossover cable and straight cable



Result :

Thus, the different types of network cables and ethernet cross-over cable / straight cable were studied.